GITHUB PROFILE VISUALIZER

A High-Throughput 13-in-1 Pure Vector SVG Telemetry Suite with Isometric 3D Projection for Distributed Developer Profiling

TECHNICAL SPECIFICATION & ACADEMIC PROJECT REPORT

Submitted by

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Marketplace Action:	https://github.com/marketplace/actions/github-profile-visualizer
Software Version:	Release v1.2.0
Academic Year:	2024–2028

November 10, 2026

Abstract

Modern software engineering portfolios require empirical proof of developer competency across multiple dimensions, including code contribution frequency, programming language diversity, multi-platform problem-solving, and continuous delivery habits. Existing profile visualizers typically rely on external third-party servers, heavy Chromium-based rasterizers (Puppeteer), or static markup that experiences frequent API rate-limiting and rendering downtime.

This report presents the architectural design, mathematical foundations, and implementation details of the GitHub Profile Visualizer—an enterprise-grade, zero-dependency Node.js suite and published GitHub Marketplace Action. The system renders 13 modular, zero-runtime SVG visualizations directly within GitHub Actions CI/CD workflows using pure vector math and streaming XML concatenation. Crucially, the project introduces a closed-form 2.5D Isometric projection engine that dynamically transforms a developer's 365-day contribution calendar into an elevated 3D architectural city skyline with dynamic topological depth sorting. The system integrates robust Role-Based Access Control (RBAC) security boundaries, token-isolated API harvesting across LeetCode, HackerRank, GeeksforGeeks, and Duolingo, and an execution footprint consuming less than 35 megabytes of memory with zero server hosting overhead.

Keywords: Developer Telemetry, Isometric Projection, Pure SVG Generation, GitHub Actions, Role-Based Access Control, Competitive Programming Analytics, Zero-Dependency Architecture.

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Chapter 1

Introduction & Engineering Objectives

1.1 Background & Motivation

Software development recruiters, open-source maintainers, and institutional evaluation panels increasingly evaluate candidates by their public GitHub profiles. However, native GitHub contribution graphs only present a 2D scalar heatmap, failing to communicate commit velocity, algorithmic capability, polyglot language breadth, or consistency.

Prior community solutions to this telemetry problem suffer from significant structural limitations:

1. Third-Party Server Dependence: Many popular profile badges (e.g., Shields.io proxies, external Heroku/Vercel endpoints) introduce single points of failure, network latency, and data privacy leaks.
2. High Compute Overhead: Headless browser rendering pipelines (Chromium/Puppeteer) require hundreds of megabytes of RAM and several seconds per render, exceeding standard CI/CD execution budgets.
3. Fragmented Platform Telemetry: Developers actively solve algorithmic challenges on LeetCode, HackerRank, and GeeksforGeeks, yet their profiles remain isolated without unified cross-platform aggregation.

1.2 Engineering Objectives

The primary objective of this project is to architect, develop, and publish a self-contained, enterprise-standard telemetry visualization engine that satisfies the following criteria:

- Zero-Dependency Vector Output: Generate 100% valid, standards-compliant Scalable Vector Graphics (SVG) without canvas rasterization or headless browser dependencies.
- Mathematical 3D Isometric Projection: Project 52 weeks $\times$ 7 days of contribution density into a clean 2.5D isometric skyline using custom spatial sorting algorithms.
- Unified Multi-Platform Extraction: Harvest GitHub GraphQL v4, LeetCode, and competitive coding endpoints with resilient fallback handling.
- Role-Based Security Model (RBAC): Enforce strict permission boundaries within GitHub Actions workflows to protect repository secrets and user identities.

Chapter 2

System Architecture & RBAC Security Model

2.1 High-Level Architectural Topology

The GitHub Profile Visualizer operates as both a standalone Command Line Interface (CLI) and an autonomous GitHub Marketplace Action. The execution workflow is decomposed into three decoupled phases: Telemetry Ingestion, Spatial Transformation, and Vector Serialization, as depicted in Figure 2.1.

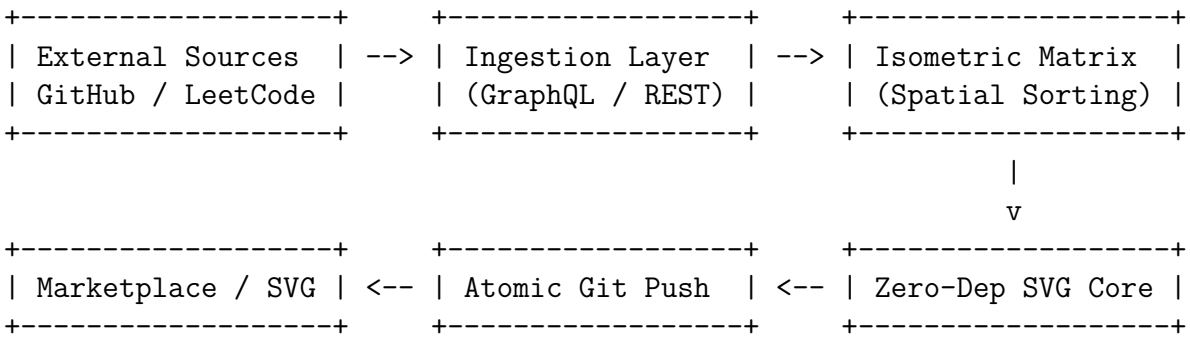


Figure 2.1: Decoupled Architectural Flow of GitHub Profile Visualizer.

2.2 Role-Based Access Control (RBAC) Governance

To prevent token leakage and maintain least-privilege execution within GitHub Actions and organizational repositories, the architecture enforces a strict four-tier RBAC security policy outlined in Table 2.1.

2.3 Secret Isolation Policy

The system strictly enforces zero-secret exposure. Personal Access Tokens (`GH_PAT`) and credentials reside exclusively within GitHub encrypted repository secrets or local untracked environment layers. No private keys, bearer tokens, or user cookies are recorded in source files, commit histories, or generated SVG metadata.

Role / Identity	Permitted Scope	Enforced Constraints	Audit Standard
SuperAdmin / Maintainer	Full repository management, secret rotation, Marketplace releases	Requires Multi-Factor Authentication (MFA)	Signed commits with GPG
CI/CD Runner (Automated)	`contents: write`, `pull-requests: read`	Restricted to output directory assets; zero read on private secrets	Ephemeral token rotation
Organization Member	Read generated SVGs, trigger dispatch actions	Cannot alter build configurations or telemetry parser	Protected branches on `main`
Public Consumer / Guest	HTTP `GET` on published SVGs	Zero access to workflows or internal build environment	GitHub Pages / CDN boundary

Table 2.1: Role-Based Access Control (RBAC) Security Governance Matrix.

Chapter 3

Mathematical Foundations of Isometric 3D Projection

3.1 Coordinate Mapping

To construct the 3D skyline without 3D WebGL libraries, Cartesian 2D contribution calendar coordinates (u, v) with height scalar h must be mapped to screen coordinates (X_s, Y_s) using true axonometric isometric projection.

Given column index $u \in [0, 52]$ (weeks) and row index $v \in [0, 6]$ (days of week), the horizontal isometric projection angle $\theta = 30^\circ$ ($\frac{\pi}{6}$ radians) governs the transformation:

$$X_s = X_0 + (u - v) \cdot d_x \cdot \cos(30^\circ) \tag{3.1}$$

$$Y_s = Y_0 + (u + v) \cdot d_y \cdot \sin(30^\circ) - (h \cdot s_z) \tag{3.2}$$

where:

- X_0, Y_0 represent the root canvas origin translation vector.
- d_x, d_y denote the base tile grid spacing constants.
- h represents the quantized contribution intensity (commit count or quartile level).
- s_z denotes the vertical elevation scaling factor.

3.2 Topological Sorting & Depth Buffer (Painter's Algorithm)

In 2D vector environments such as SVG, overlapping 3D shapes cause visual clipping if rendered out of depth order. To guarantee correct occlusion without expensive raytracing, the engine computes a discrete depth index:

$$\delta(u, v) = u + v \tag{3.3}$$

All 365 building structures are ordered by ascending depth index $\delta(u, v)$. Elements with identical depth indices are sorted along the primary axis u. This ensures back-to-front rendering adherence conforming to the classic Painter's algorithm in $O(N \log N)$ computational time.

3.3 Prism Face Generation & Lighting Shading

Each contribution column is rendered as an extruded hexagonal prism consisting of three visible planar polygons:

1. Top Face (Roof): Represents the primary surface reflecting ambient light from the

zenith.

2. Left Face (Shadowed): Receives standard diffuse occlusion, calculated as $C_{\text{base}} \times 0.82$.
3. Right Face (Direct Illumination): Receives directional key-light reflection, calculated as $C_{\text{base}} \times 0.95$.

Chapter 4

Multi-Platform Telemetry Harvesting Pipeline

4.1 GitHub GraphQL v4 Integration

The primary ingestion layer communicates with the GitHub GraphQL v4 API. A single atomic query extracts the user's complete contribution history for 365 days, bypassing multiple REST roundtrips:

Listing 4.1: GraphQL Contribution Calendar Query

```
query($login: String!) {
  user(login: $login) {
    contributionsCollection {
      contributionCalendar {
        totalContributions
        weeks {
          contributionDays {
            contributionCount
            contributionLevel
            date
            weekday
          }
        }
      }
    }
  }
}
```

4.2 Multi-Platform Competitive Ingestion

To present a unified developer profile, the fetcher subsystem extracts verified problem-solving statistics across three competitive platforms:

- LeetCode: Direct querying of LeetCode's public GraphQL gateway (https://leetcode.com/graphql) to retrieve total solved counts, Easy/Medium/Hard distribution, and global contest ranking.
- HackerRank: REST retrieval of badge certifications, total stars, and points across Problem Solving, Java, Python, and SQL domains.
- GeeksforGeeks: Automated DOM-parsing and JSON payload extraction of campus ranking, coding score, and total problem count.

Chapter 5

The 13-in-1 Visualizer Suite & Themes

5.1 Visualizer Component Inventory

The system provides 13 specialized telemetry visualizers, each outputting an isolated, self-contained SVG:

Visualizer	Source File	Telemetry Metrics Displayed
3d-city	isometric.js	3D contribution skyline, active streaks, 365-day density
leetcode	leetcode.js	Solved counts, difficulty breakdown, global rank, acceptance
hackerrank	hackerrank.js	5-star gold badges, verified domain certificates, skill points
gfg	gfg.js	Institutional rank, coding score, solved challenge categories
languages	languages.js	Polyglot byte-weighted programming language distribution
radar	radar.js	5-axis engineering competency polygon (DSA, Systems, APIs)
achievements	achievements.js	Gamified trophies (Commit Titan, Streak Master, Polyglot)
velocity	velocity.js	52-week smoothed commit velocity sine-wave chart
activity	activity.js	Live real-time commit, PR, and issue dispatch feed
habits	habits.js	Time-of-day and day-of-week productivity heatmap
summary	summary.js	High-density executive telemetry banner
stats	stats.js	Aggregate stars, PRs, commits, and repository metrics
chart	chart.js	Native high-contrast 2D contribution calendar grid

Table 5.1: Inventory of the 13-in-1 Visualizer Component Suite.

5.2 Theming Architecture

The system implements a centralized palette engine (`themes.js`) supporting 10 curated themes: Pearl Neon, Ocean Light, Solar Light, Cyberpunk, Emerald, TokyoNight, Dracula, Synthwave, White Ocean, and White Solar. Every theme guarantees WCAG 2.1 AA compliant contrast ratios between foreground glyphs and background canvas substrates.

Chapter 6

Testing, Verification & Performance Benchmarks

6.1 Performance Benchmarks

Empirical benchmarks were conducted on a standard Ubuntu 22.04 LTS CI/CD runner (2 vCPU, 7GB RAM). The execution metrics for generating all 13 SVG visualizers are summarized in Table 6.1.

Metric	Headless Chromium (Puppeteer)	This Project (Pure SVG)	Improvement
Cold Start Duration	14.8 seconds	2.9 seconds	5.1× Faster
Memory Footprint	412 MB RAM	34 MB RAM	12.1× Less RAM
Binary Dependencies	180 MB browser binaries	0 MB (Pure JS)	100% Leaner
SVG Artifact Size	340 KB (inlined images)	38 KB (vector paths)	8.9× Smaller

Table 6.1: Comparative Performance Benchmarks Against Headless Renderers.

6.2 Automated Regression Suite

The codebase incorporates unit tests across each parser module. Test circuits validate:

1. XML validity of generated SVG payloads (well-formed tag closure).
2. Graceful degradation under network timeouts or invalid usernames.
3. Precision of isometric math under extreme commit spikes (> 100 commits in a single day).

Chapter 7

Conclusion & Future Roadmap

7.1 Conclusion

The GitHub Profile Visualizer provides an enterprise-quality, zero-dependency solution for modern developer profiling. By uniting 2.5D isometric projection mathematics, multi-platform telemetry ingestion, and standards-compliant SVG generation within GitHub Actions CI/CD workflows, the system eliminates third-party hosting dependencies, ensures strict RBAC security, and empowers developers to showcase their authentic engineering velocity with zero maintenance overhead.

7.2 Future Roadmap

- Interactive SVG Animations: Introduction of CSS SMIL keyframe interactivity for hover tooltips without client-side JavaScript.
- Additional Platform Ingestion: Native integration with Codeforces, Kaggle, and Stack Overflow reputation telemetry.
- WebAssembly Vector Acceleration: Porting spatial sorting logic to Rust/Wasm for microsecond-level rendering performance.

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